Standard Operating Procedure: Centrifugal Pump P-101

Document Ref: SOP-OPS-P101-01 | Classification: Restrictive

1. Pre-Start Inspections

Before executing a startup command for the P-101 main suction pump, operators must verify the following pre-start checks:

- Check lubrication oil levels in the main bearing housing. Level must be between 50% and 75% on the sight glass indicator.

- Confirm that the suction gate valve V-101A is fully open. WARNING: Running the pump with a closed suction line will cause dry-run cavitation.

- Inspect the mechanical seals for any visible leakage. A drip rate exceeding 10 drops per minute requires seal inspection.

- Ensure cooling water flow is active to the seal flush line with a minimum flow rate of 2.5 GPM.

2. Startup Sequence

Follow these sequential steps to safely start Pump P-101:

Step 1: Rotate the pump shaft manually by hand to ensure the impeller is free of obstructions.

Step 2: Fully close the discharge globe valve V-102B. (Starting centrifugal pumps against a closed valve reduces current surge load).

Step 3: Turn on the main electrical isolator panel switch in MCC-01 cabinet.

Step 4: Press the START button on the local control interface panel.

Step 5: Monitor the motor current draw. The startup peak current should drop to the normal operating load of 45 Amps within 3 seconds.

Step 6: Slowly crack open the discharge valve V-102B until the system pressure stabilizes at 6.2 bar operating pressure.

3. Operational Specifications

Parameter	Normal Value	Trip Limit
Discharge Pressure	6.2 bar	7.5 bar
Motor Temperature	65 C	85 C
Vibration Velocity	2.8 mm/s	4.5 mm/s
Flow Rate	120 GPM	150 GPM

4. Health & Safety Compliance

This equipment processes hazardous hydrocarbons. Lockout/Tagout (LOTO) procedures must be enforced prior to any intervention. Minimum required PPE includes hard hat, steel-toed boots, protective safety goggles, and anti-static overalls.